ROGERS STATE UNIVERSITY **Coding and Smart Systems Camp**

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Blinking LED

Hardware Components Required:

- Raspberry Pi 4 Model B with GPIO extension
- Breadboard
- 1 LED
- 1 Resistor (220 Ω)
- 1 Yellow Wire
- 1 Black Wire

Step-by-step instructions to connect hardware components:

1. Plug the LED on the breadboard so the long leg (anode) and short leg (cathode) are in different rows.

2. Plug one end of the 220 Ω resistor into the same row as the LED long leg. Plug the other end of the resistor into an empty row (resistor row).

3. Take the yellow wire. Plug one end into the resistor row. Plug the other end into GPIO 17 (pin 11) row on the breadboard.

4. Take the black wire. Plug one end into the row with the LED short leg. Plug the other end into GPIO GND (pin 6) row on the breadboard.

5. Power on the Pi by turning on the black switch.

6. On the screen, **click on the projects folder** and then click on **Day1** folder.

7. Click on the **project1 folder** and then **click on the blink.py** file.

8. Program the LED to blink.

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Python code (Solution):

```python
# blink_led.py

from gpiozero import LED
from time import sleep

led = LED(17)

try:
    while True:
        led.on()
        sleep(0.5)
        led.off()
        sleep(0.5)

except KeyboardInterrupt:
    pass

finally:
    led.off()
    print("LED stopped. Program terminated.")

```